

Bias in Federated Learning: A Comprehensive Survey

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Federated Learning (FL) enables collaborative model training over multiple clients' data, without sharing these data for better privacy. Addressing bias in FL remains a challenge. In this article, we first present a taxonomy of FL bias, presenting the causes and the different types of FL bias, namely demographic bias, performance-related bias, and contribution-related bias. We then categorize FL bias mitigation, in terms of used methods and provided guarantees, before providing a comprehensive and comparative analysis of existing works. Finally, we highlight key challenges and open research directions, including the impact of FL bias mitigation on model utility, privacy, and robustness.

CCS Concepts: • **Computing methodologies** → **Distributed computing methodologies; Machine learning; Distributed artificial intelligence**; • **Social and professional topics** → *User characteristics*;

Additional Key Words and Phrases: Federated learning, bias, fairness

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1 Introduction

The rapid growth in computer software and hardware technologies has made a huge amount of data accessible. However, centralizing this data comes with high costs, time-consuming process, and privacy concerns. In response to these limitations, **Federated Learning (FL)** has emerged as a prominent approach in critical areas such as healthcare [99, 109], loan granting [125], and IoT systems [81]. Despite its advantage in preserving data privacy by allowing FL clients to collaboratively train models without sharing their local data [93], FL remains inherently prone to unfairness and bias. One major cause is data imbalance, often leading to biased models. Additionally, FL client selection and FL model aggregation algorithms may favor certain FL clients, inadvertently skewing contributions, excluding underrepresented clients, and reinforcing disparities. These factors introduce systemic bias into the FL model, ultimately impacting its fairness across different demographic groups and FL participants.

Bias represents a widely recognized concern that has been extensively explored in **Machine Learning (ML)** [22, 94]. It primarily arises from unbalanced, incomplete, flawed, or detrimental

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models and datasets, resulting in unfair model predictions that favor one demographic group over another based on sensitive attributes such as religion, race, gender, and age, which are legally, all protected by law [35, 145]. One notorious example is Amazon’s recruiting tool which favored men, because women were underrepresented in the training dataset and the resulting model was biased in favor of the over-represented group [28]. The severity of bias in **Artificial Intelligence (AI)** has motivated several legislative actions, such as the National AI Initiative Act in the US [131], and the EU AI Act [130]. In FL, bias can manifest in various forms, with demographic bias being a notable concern. This arises when FL clients train models using locally biased data, propagating this bias to the globally aggregated model [17]. Furthermore, the distributed and collaborative nature of FL introduces two additional bias types beyond demographic bias, namely performance-related bias and contribution-related bias. Performance-related bias centers on disparities among clients performance in the federated network. This bias type captures the variations in individual clients’ model performance, highlighting the need for a more uniform model performance. For instance, in case of FL clients that represent hospitals, all clients, whether being large or small hospitals, should receive a model with the same prediction quality. In contrast, in some cases, there is the need to ensure that each client receives a specialized model which performance is proportional to its contribution to the overall federated learning process. When this is not the case, contribution-related bias arises, reflecting the imbalance between model quality and the varying contribution levels of FL clients. This is, for instance, the case of mobile FL clients involved in a distributed geolocation-based application, where clients learning with larger location datasets receive models with higher prediction quality. Thus, mitigating performance-related bias and contribution-related bias involves opposing goals. Indeed, addressing performance-related bias seeks to equalize model performance across clients independently from their contribution, whereas mitigating contribution-related bias aims at providing to each client a specialized model with performance that is proportional to its contribution. Finally, these multifaceted FL bias types necessitate careful consideration and targeted mitigation strategies to ensure fair and unbiased learning.

In classical centralized ML frameworks, extensive exploration has been conducted on bias mitigation [10, 15, 22, 94, 165]. However, the conventional solutions often assume unrestricted access to the data, a premise that contradicts the FL privacy constraints. Moreover, applying these centralized methods locally is not only costly expensive but also does not guarantee a global demographic bias mitigation, given the significant data heterogeneity characterizing FL environments. In FL, diverse bias mitigation methods have emerged to address the different types of bias encountered. These include strategies to mitigate demographic bias [26, 38, 150, 160], which strives to equalize model performance across different demographic groups, ensuring a fair and unbiased representation for all. Additionally, there are approaches dedicated to mitigate performance-related bias [44, 79, 97], aiming at achieving uniform model performance among FL clients, thereby promoting equality in the learning process. Other methods tackling contribution-related bias have emerged, ensuring that each FL client receives a model tailored to its unique contribution [90], fostering a more personalized and fair learning experience in the federated ecosystem. As FL continues to gain prominence, refining and expanding upon these bias mitigation strategies remains a critical aspect.

Our survey delves into the three types of FL bias, their underlying causes, and an analysis of the strategies employed to tackle this issue. In essence, our survey provides a holistic understanding of the intricacies surrounding FL bias. Through this comprehensive exploration, we aim at offering valuable insights into the evolving landscape of FL bias mitigation, providing a roadmap to navigate the bias challenges in FL systems.

Table 1. Review of Surveys on Bias in FL. With Different Types of Bias (Demographic bias - Dem., Performance-related bias- Perf., Contribution-related bias- Cont.), bias mitigation guarantees (Minimized bias - Min., Constrained bias - Cons.), heterogeneity and scalability (FL data heterogeneity - Heter., Scalability with regard to data - ScalD., Scalability with regard to the FL clients number - ScalC.), evaluation metrics (Bias, Contribution, Quality, Cost)

Survey	Types of FL bias			Bias mitigation guarantees		Heterogeneity and Scalability			Evaluation metrics				#Articles analyzed
	Dem.	Perf.	Cont.	Min.	Cons.	Heter.	ScalD.	ScalC.	Bias	Contribution	Quality	Cost	
[134]	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	24
[18]	✓	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	14
[124]	✓	✓	✓	✗	✗	✗	✗	✗	✓	✓	✓	✗	20
[115]	✓	✓	✓	✗	✗	✗	✗	✗	✓	✓	✓	✗	28
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	66

1.1 Related Surveys

While numerous works have explored bias and fairness in ML [9, 47, 57, 67, 94, 140], there remains a lack of a comprehensive survey specifically addressing bias in FL. Despite the availability of surveys on FL [75, 159, 168], only a few delve into the nuanced aspects of bias in FL settings. Table 1 shows existing surveys on bias in FL. Both works in [134] and [18] lack the coverage of the three FL bias types, and overlook critical dimensions of FL bias mitigation, which involve minimizing bias as much as possible or ensuring it does not exceed a given threshold. Additionally, these works fail to consider crucial factors like evaluation metrics, and the scalability of bias mitigation methods. Studies in [115, 124] explore FL bias types more broadly. However, both overlook critical aspects such as bias guarantees, and the scalability of bias mitigation methods. Our survey fills this critical research gap by conducting an in-depth investigation into FL bias across 66 selected articles. We provide a holistic analysis of the different types of FL bias, their underlying causes, and the array of strategies employed for their mitigation. Through this extensive exploration, our work aims at delivering a roadmap for addressing bias challenges in FL systems, while highlighting future research directions.

Our literature review follows the PRISMA guidelines [104]. We conducted an extensive search of articles published up to January 2025 across academic databases, including ACM Digital Library, IEEE Xplore, Google Scholar, Elsevier, Semantic Scholar, and DBLP, focusing on works published in peer-reviewed journals (i.e., journals ranked Q1 or Q2 on SJR), and top-tier conferences (i.e., conferences ranked A*, A, or B in the CORE ranking portal). Articles were identified using search terms such as “Bias”, “Bias Mitigation”, “Fairness” and “Federated Learning” refined with Boolean operators (e.g., AND, OR) to maximize the comprehensiveness of our search queries. The search involved multiple screening rounds, including title and abstract reviews, full-text analysis, and cross-referencing bibliographies from key studies to identify additional relevant articles. This systematic approach ensures a thorough and up-to-date representation of bias-related research in FL.

1.2 Research Contributions

Recognizing the critical need to address the existing research gap, we survey recently published articles on FL bias from top-tier conferences and journals. In particular, we not only investigate the three types of FL bias but also highlight their nuanced aspects, underlying their causes, and the methods employed for effective mitigation. Through a rigorous and in-depth analysis, our survey seeks to provide valuable insights into the intricacies of bias in FL. Our major contributions can be summarized as follows:

- A comprehensive investigation of all three types of FL bias, including demographic bias, performance-related bias and contribution-related bias.
- An in-depth analysis of nuanced aspects and underlying causes of each bias type.
- An introduction of a new taxonomy to classify existing bias mitigation systems.

- A review of a diverse array of bias mitigation methods, analyzing the bias guarantee of each bias mitigation method, distinguishing between minimizing bias and maintaining it below a predefined threshold.
- Consideration of multiple evaluation metrics, including those measuring bias, impact on model utility, and cost metrics.
- A review of methods that provide theoretical guarantees and those with empirical evaluation. For the latter, we consider the scalability with regard to the number of client and the scalability with regard to the training data size, as well as, the robustness to data heterogeneity in **Non-Independent and Identically Distributed (non-IID)** settings.
- An inclusion of both data partitioning, FL system type and FL architecture perspectives.
- A prospective discussion on open research directions

Our survey strives to develop a thorough understanding of FL bias. Within this scope, we uncover several avenues for further research, such as the automated identification of the attributes that induce demographic bias in FL, investigating the intersection of FL bias types and proposing comprehensive strategies for their collective resolution. Additionally, we extend the open research directions to the bias in multimodal FL, an evolving paradigm that leverages diverse data modalities from different devices. This comprehensive review not only provides insights into the current state of FL bias research but also outlines promising directions for future investigations.

1.3 Article Roadmap

In the rest of the article, we first give an overview of FL in Section 2, providing essential background on the FL paradigm. In Section 3, we introduce a taxonomy of FL bias, defining each bias type in FL, its causes, and relevant evaluation metrics. Next, Section 4 presents real-world use cases, illustrating how bias manifests in practical FL applications. In Section 5, we outline a taxonomy of FL bias mitigation, namely bias mitigation guarantees, methods, and the metrics used to evaluate the effectiveness of FL bias mitigation. We then provide a systematic review of the existing literature on FL bias mitigation in Section 6. Finally, we discuss the potential avenues for further research in Section 7, before concluding in Section 8.

2 Background on Federated Learning

In this section, we introduce the fundamentals of FL, including its definition, system architecture, and data partitioning schemes to support the discussions that follow.

2.1 Overview of Federated Learning

FL is a distributed learning paradigm that enables multiple clients $C_1, C_2, \dots, C_N$, each one having respectively a local dataset $D_1, D_2, \dots, D_N$, to collaboratively train a global model θ over these distributed datasets without sharing clients' data [93]. FL clients' datasets are known to be non-IID [77]. The overall FL protocol is orchestrated by a server through successive FL learning rounds. At each FL round, the server selects a subset of k clients that will participate in that round. Each client C_i trains a local model θ_i on its local data D_i , and then shares its local model's parameters with the server. The latter aggregates clients' model updates using, for instance, the classical federated averaging algorithm FedAvg [93], or any other FL aggregation method [64, 78]. Thus, in case of the FedAvg aggregation method, the overall FL model is produced following Equation (1). Finally, the resulting global FL model is shared by the server with the clients.

$$\theta = \sum_{i=1}^{i=k} \frac{|D_i|}{|D_1 \cup \dots \cup D_k|} \cdot \theta_i. \quad (1)$$

2.2 Federated Learning Architecture

There are two main FL system architectures, namely client-server FL architecture, and decentralized FL, as described in the following.

2.2.1 Client-server Federated Learning. Client-server FL is a distributed system where a central server orchestrates a collaborative learning involving several FL clients, in order to train a global model while maintaining client data private. Each FL round involves three main steps, which are repeated iteratively as follows. First, the server initializes the global model parameters, selects clients to participate in the FL round, and sends the global model parameters to the selected clients. Then, FL clients train their local models on their private training data, and then send their local model updates to the server. Finally, the server aggregates the received clients' model updates, using for instance weighted averaging, to produce an updated version of the global model. This global model is then sent back to the selected FL clients. This process is repeated at each FL round.

2.2.2 Decentralized Federated Learning. In decentralized FL, each client locally trains a ML model using its own data, and incorporates updates from other neighbor clients. The updated model information is then shared with other FL clients. There are two main schemes of decentralized FL, namely cyclic transfer and random transfer [21, 112, 154]. In cyclic transfer, clients are organized in a circular chain. A client receives model information from its preceding client, updates the received model using its local data, then sends the updated model to its following client. Whereas in random transfer, a client selects another client randomly, and exchanges updated model information. There are also decentralized FL systems that are based on the blockchain technology [59, 74], to enhance security, transparency, and trust in collaborative training. In such systems, the blockchain acts as a decentralized ledger, securely recording model updates and transactions to validate and store updates across clients, ensuring data integrity and enabling traceability without relying on a central authority [51, 114]. Designing an effective communication for a decentralized FL architecture is a complex task that requires careful consideration to handle communication overheads [8, 82].

2.3 Federated Learning Data Partitioning

Data are distributed among FL clients in two main ways, namely horizontally or vertically FL [148], as discussed below.

2.3.1 Horizontal Federated Learning. **Horizontal FL (HFL)** refers to a FL system where datasets from different clients have the same feature space, but distinct data records. For instance, in case of a HFL system that consists of hospitals that collaborate for disease diagnosis, the feature space describing patient information is the same in all hospitals, but the patient records are different among the hospitals.

2.3.2 Vertical Federated Learning. **Vertical FL (VFL)** is used when FL clients overlap in the sample space but differ in the feature space [86]. For instance, in a VFL system that involves a banking service client and an e-commerce client, the bank provides data on users' revenue, expenditure, and credit ratings, while the e-commerce company shares users' browsing and purchasing histories. By combining their distinct information, they develop a prediction model for product purchase recommendations based on comprehensive user and product information, enhancing predictive accuracy while maintaining data privacy.

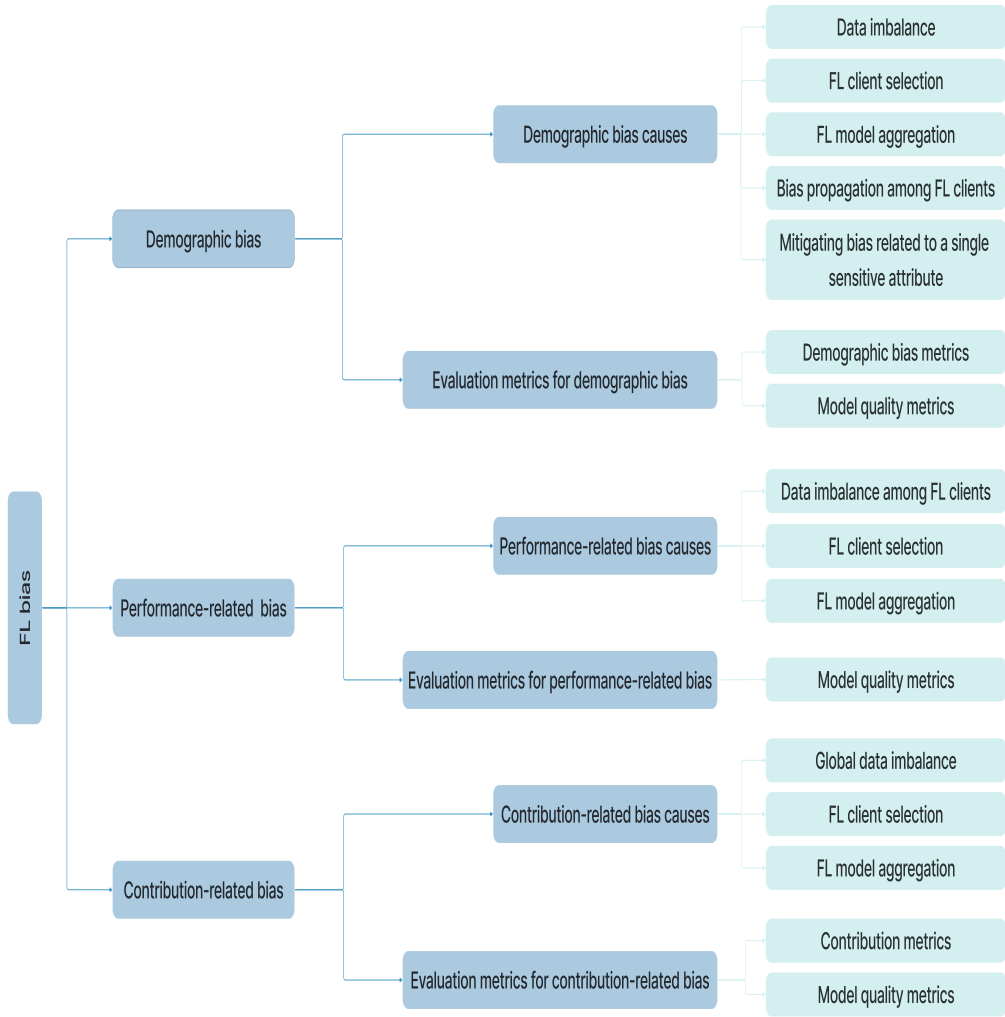


Fig. 1. Taxonomy of bias in FL.

3 Bias in Federated Learning: A Taxonomy

This section provides an overview of bias in FL and introduces a taxonomy categorizing it into three distinct types. We start with a general overview of FL bias, before detailing each type of FL bias, its causes, and its evaluation metrics. Figure 1 illustrates this taxonomy, showing the three bias types in FL along with their associated causes and evaluation metrics.

3.1 Overview of Bias

In AI research, the definition of bias varies across research areas, reflecting distinct perspectives on its manifestation and impact. In responsible AI, bias refers to systematic imbalances in algorithmic outcomes, resulting in unfair advantages or disadvantages for groups often defined by sensitive attributes such as gender or ethnicity [56]. Building on this, trustworthy AI emphasizes fairness as a foundation for transparency, explainability, and trust, framing bias as a barrier to building systems that behave ethically and predictably [66]. Whereas reliable AI views bias as

inconsistencies in model performance across varying conditions, undermining the predictability and stability of AI systems [119]. Lastly, in robust AI, bias is seen as a vulnerability, increasing susceptibility to adversarial attacks and unexpected performance degradation, particularly affecting underrepresented groups [43].

Despite these varying perspectives, bias across these research areas consistently manifests as systematic disparities, resulting in unfair model behavior across different groups and environments. In this article, we define bias in FL as systematic disparities in model outcomes that manifest in three primary dimensions, disparities across demographic groups (i.e., demographic bias), inconsistencies in performance across FL clients (i.e., performance-related bias), and misalignment between clients' contributions and the benefits they derive from the global model (i.e., contribution-related bias). In the following, we will outline the definitions, causes, and evaluation metrics associated with each bias type.

Despite these varying perspectives, bias across these research areas consistently manifests as systematic disparities, resulting in unfair model behavior across different groups and environments. In this article, we define bias in FL as systematic disparities in model outcomes that manifest in three primary dimensions, disparities across demographic groups (i.e., demographic bias), inconsistencies in performance across FL clients (i.e., performance-related bias), and misalignment between clients' contributions and the benefits they derive from the global model (i.e., contribution-related bias). We will see later that the mitigation of the two latter types of FL bias, namely performance-related bias and contribution-related bias, involves opposing goals. In the following, we will outline the definitions, causes, and evaluation metrics associated with each bias type.

3.2 Demographic Bias

In the following, we first define demographic bias in FL, before presenting its causes, and the evaluation metrics to measure this type of bias and its side effects on other aspects, such as model quality.

3.2.1 Definition of Demographic Bias. Demographic bias is defined as the presence of favoritism or discrimination that unequally affects some demographic groups based on their sensitive attributes. The sensitive attributes cover demographic characteristics that have the potential to lead to discriminatory behavior, such as gender, race, ethnicity, age, and so on [35, 145], and which use in decision-making is prohibited by law [37, 122, 132, 133]. In FL, the presence of data sensitive attributes introduces the risk of bias and discrimination, as models may inadvertently learn and perpetuate unfair patterns among the different demographic groups. Formally, we consider a set of sensitive attributes $S_A = \{S_1, S_2, \dots, S_{s_a}\}$. For simplicity sake, we assume that a sensitive attribute has a binary value $S_j \in \{v_1^j, v_2^j\}$, to represent a privileged and unprivileged demographic groups, data associated to these groups is denoted as $D_{S_j=v_1^j}$, and $D_{S_j=v_2^j}$. Additionally, we considered a learning model θ for a binary prediction task with the data label $y \in \{0, 1\}$. The model outcome variable $\hat{Y} \in \{0, 1\}$ has two values. Value 1 is referred to as the model's positive outcome, while 0 is referred to as the negative outcome. Here, $Perf(\theta, D_{S_j=v_1^j}, \hat{Y}=y)$ represents the model performance for the privileged group (e.g., model accuracy), while $Perf(\theta, D_{S_j=v_2^j}, \hat{Y}=y)$ denotes the performance for the unprivileged group. Thus, demographic bias exists if:

$$\exists S_j \in S_A, \quad \left| Perf(\theta, D_{S_j=v_1^j}) - Perf(\theta, D_{S_j=v_2^j}) \right| > \epsilon.$$

where ϵ is a predefined threshold representing an acceptable difference in model performance between the two demographic groups. This definition emphasizes that demographic bias arises when performance disparities between groups exceed an acceptable margin. These disparities translate

into real-world consequences, leading to unfair models that exhibit discriminatory behavior and unfair treatment, such as sexist or racial segregation. For instance in [127], it has been demonstrated that some health sensors, such as oximeters, tend to misclassify individuals of color as they have primarily been tested on white populations. Likewise, voice recognition models may exhibit greater accuracy for men than for women due to the variations in vocal characteristics between genders. Indeed, women's voices usually have a higher pitch and thinner quality, which presents challenges in capturing voice features compared to men's voices [129].

3.2.2 Causes of Demographic Bias. Demographic bias in FL can be attributed to several factors, which we detail in the following.

Data Imbalance. In FL, data distribution among clients is known to be non-IID [77, 78, 121]. When related to sensitive attributes, this property refers to variations in their distribution and how they relate to other attributes and labels across clients, leading to demographic disparities. These disparities are a key source of bias in FL models, typically manifesting either globally or locally. Global data imbalance occurs when the union of all FL clients datasets is imbalanced, leading to a global model that performs inequitably across demographic groups. Local data imbalance, on the other hand, arises when individual clients have imbalanced datasets. Even if the combined dataset across all clients is balanced, local imbalance can still introduce bias. For instance, bias may occur if the FL process inadvertently favors clients with underrepresented demographic groups through selection or aggregation methods.

Bias Caused by FL Client Selection. At each FL round, the server selects a subset of FL clients that will participate to that round, based on several factors such as network communication times [101], model training times [68], or energy consumption [157]. The client selection policy is critical to a FL system in terms of training efficiency, final model's quality and fairness [7, 40]. When the server repeatedly selects a specific type of clients, the selection policy may have undesirable side-effects by neutralizing some portions of data, reducing data diversity and inducing demographic bias [53]. Specifically, if clients with underrepresented demographic groups are consistently excluded from selection, their data remains unrepresented in the training process, resulting in a biased FL model.

Bias Caused by FL Aggregation. In order to produce the global FL model, FL clients' local model updates are aggregated through a weighted average, where the weights reflect the relative amount of data contributed by each client, the client's computational power, or communication capacity [3, 69, 78, 93]. The underlying FL aggregation method may lead to an unfair model [95]. Notably, when clients with biased data are given higher weights than clients with non-biased data.

Bias Propagation among FL Clients. The presence of imbalanced, flawed or biased data in a FL client may lead to the propagation of the demographic bias across other FL clients [17], including those with fair data. Indeed, in a FL system, even clients with unbiased local data, which would otherwise enable them to train fair models, can inherit a biased global model due to other clients with biased local data. The bias exhibited by few clients toward a particular demographic group, related to a sensitive attribute such as gender or race, has the ability to spread throughout other participating FL clients.

Side-effects of Mitigating Bias Related to a Single Sensitive Attribute. FL clients' data may embed multiple sensitive attributes such as gender, race, age, and so on, and solely considering a single sensitive attribute for bias mitigation is insufficient [12]. In the presence of multiple sensitive attributes, applying bias mitigation with regard to a single sensitive attribute may introduce or exacerbate bias with respect to other sensitive attributes present in the data, even if the model was

not initially biased with respect to these attributes. While attempts have been made to address bias in FL, most existing works focus on mitigating bias related to a single sensitive attribute. However, handling bias related to a single attribute may inadvertently amplify bias with respect to other attributes [150]. For instance, if a re-weighting technique is applied to mitigate bias related to a given sensitive attribute, it may have as a side-effect lower weights relatively to other sensitive attributes, thereby, introducing more bias for the unconsidered attributes.

3.2.3 Evaluation Metrics of Demographic Bias. In the following, we present demographic bias metrics, which focus on quantifying disparities in predictions and outcomes across demographic groups, providing a measurable representation of bias. Additionally, we introduce model quality metrics, used to evaluate the model's performance across different demographic groups, highlighting potential inequalities.

Demographic Bias Metrics. Demographic bias is measured using dedicated tools. In this section, we introduce key metrics designed to quantify disparities in predictions and outcomes across demographic groups.

Statistical Parity Difference (SPD) is used to assess the extent to which a model's predictions depend on a particular sensitive attribute [38]. A model achieves perfect demographic parity with regard to a set of sensitive attributes, if for each sensitive attribute $S_j \in S_A$, the probability of two demographic groups to have a positive outcome from the FL model is the same. Formally, SPD measures the difference between the probability to have a positive outcome for two demographic groups, as defined in Equation (2):

$$SPD_{S_j} = Pr(\hat{Y} = 1 | S_j = v_1^j) - Pr(\hat{Y} = 1 | S_j = v_2^j). \quad (2)$$

Disparate Impact (DI) occurs when the FL model outcomes $\hat{Y}$ widely differ for different demographic groups [39]. DI is measured as the ratio of the probability of positive outcomes between the privileged and unprivileged groups. Formally, it is calculated following Equation (3).

$$DI_{S_j} = \frac{Pr(\hat{Y} = 1 | S_j = v_1^j)}{Pr(\hat{Y} = 1 | S_j = v_2^j)}. \quad (3)$$

Average Odds Difference (AOD) allows to capture model prediction quality [45]. AOD considers the **true positive rates (TPR)** and the **false positive rates (FPR)** among different demographic groups, and is formally defined in Equation (4):

$$AOD_{S_j} = \frac{1}{2}[(FPR_{S_j=v_1^j} + FPR_{S_j=v_2^j}) - (TPR_{S_j=v_1^j} + TPR_{S_j=v_2^j})], \quad (4)$$

where $FPR_{S_j=v_1^j}$ and $FPR_{S_j=v_2^j}$ respectively denote the false positive rate for the privileged and unprivileged groups, while $TPR_{S_j=v_1^j}$ and $TPR_{S_j=v_2^j}$ denote the true positive rate for the two groups.

Equal Opportunity Difference (EOD) considers the true positive predictions [45]. Formally, the FL model prediction is considered fair if the true positive rate remains unaffected by the sensitive attributes. EOD metric is calculated in Equation (5):

$$EOD_{S_j} = Pr(\hat{Y} = 1 | S_j = v_1^j, Y = 1) - Pr(\hat{Y} = 1 | S_j = v_2^j, Y = 1). \quad (5)$$

Discrimination Index (DiscI) measures the disparity in the F1-score between the demographic groups [160]. A high discrimination index indicates that the model's prediction quality is higher

for one of the demographic groups compared to the other. $DiscI$ metric for a model θ with regard to a sensitive attribute S_j can be expressed as

$$DiscI_{S_j} = F1-score_{S_j=v_2^j}(\theta) - F1-score_{S_j=v_1^j}(\theta), \quad (6)$$

where $F1-score_{S_j=v_i^j}(\theta)$ denotes the F1-score of the model θ computed using only data records from the group v_i^j .

Model Quality Metrics. In addition to demographic bias metrics, model quality metrics provide a way to quantify performance differences across demographic groups, offering insights into how bias manifests in model outcomes. Classical metrics, such as accuracy, precision, recall, and F1-score, are commonly used for this purpose [98], the definitions of these metrics are provided below.

Accuracy is a statistical metric that measures the proportion of correct predictions made by a learning model, as expressed in Equation (7). It provides a concise evaluation of the model's performance by comparing predicted labels with true labels to assess how well a model performs in terms of its ability to correctly classify or predict data.

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \quad (7)$$

Precision indicates the proportion of true positives among all positive predictions, reflecting the model's ability to minimize false positives, as defined in Equation (8).

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}. \quad (8)$$

Recall measures the proportion of true positives identified out of all actual positives, emphasizing the model's sensitivity to detecting relevant instances, as given in Equation (9).

$$Recall = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}. \quad (9)$$

F1-score represents the harmonic mean of precision and recall, providing a balanced metric when dealing with imbalanced classes, as shown in Equation (10).

$$F1-score = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (10)$$

3.3 Performance-Related Bias

In the following, we first define performance-related bias in FL, before presenting its causes, and the evaluation metrics to measure this type of bias.

3.3.1 Definition of Performance-related Bias. Due to its decentralized nature, FL also introduces other sources of bias. A notable form of bias that emerges in this context, termed performance-related bias, focuses on the disparities in model performance observed among FL clients. Such a bias may stem from several factors, including differences in the quality or quantity of data available to each FL client [54]. Since FL clients can widely vary in terms of data sources, and the conditions under which their data were collected and processed, these discrepancies can have a significant impact on the clients' learning process. Thus, performance-related bias presents a critical challenge in FL systems, with a performance imbalance that can result in an uneven distribution of benefits and advantages among the FL clients [97]. Formally, performance-related bias emerges when

disparities exist in model performance across FL clients, with $Perf(\theta_i, D_i)$ presenting the model performance metric measured on client C_i 's local dataset D_i . This is expressed as follows:

$$\exists i, j, \quad |Perf(\theta_i, D_i) - Perf(\theta_j, D_j)| > \epsilon, \quad i \neq j$$

where ϵ is a predefined tolerance threshold for acceptable performance disparity between clients. Importantly, performance-related bias is independent of client contribution levels. Regardless of how much a client contributes to the FL training process, the resulting model should ideally deliver consistent performance across all clients. If clients experience varying model performance, performance-related bias is present, even if their contributions differ. For instance, in a FL healthcare system, the learning model can be more efficient in hospitals with larger and more comprehensive patient databases compared to the smaller healthcare providers with more limited datasets. As a consequence, the quality of medical diagnoses, treatment recommendations, or predictive models may vary significantly across the FL system. Such disparities can raise ethical concerns, as they may lead to unequal access to accurate healthcare.

3.3.2 Causes of Performance-Related Bias. Various factors contribute to performance-related bias in FL, which are outlined below.

Data Imbalance among FL Clients. Data imbalance among FL clients arises from the non-IID nature of FL, which reflects variations in data distribution across clients. These variations often stem from differences in client-specific conditions, such as data collection methods, user behaviors, or application-specific contexts. Such data imbalance poses significant challenges in aligning the global model with the diverse local distributions, resulting in disparities in the performance of local models. Specifically, the global model may overfit to dominant patterns from certain clients while underperforming on those with distinct distributions. This imbalance limits the model's ability to generalize effectively across all clients, introducing performance disparities among FL clients and ultimately leading to performance-related bias.

Bias Caused by FL Client Selection. The client selection policy is critical to a FL system in terms of training efficiency, final model's quality and fairness [7, 40]. As discussed in the causes of demographic bias (see Section 3.2.2), client selection policies may significantly impact model performance and fairness in FL systems. When the server repeatedly selects a specific type of clients, these clients contribute more to the global model evolution, and may introduce FL model bias. For instance, in [157], clients with higher computation performance are more favored. In such a case, always selecting the faster client devices can improve model training times, however, at the expense of depriving clients with low priority of the opportunity to contribute to the training. This may result in skewed models that exhibit reduced accuracy when making predictions for under-selected clients, thus, resulting in performance-related bias.

Bias Caused by FL Aggregation. As discussed in the context of demographic bias causes (see Section 3.2.2), the global FL model is typically formed as a weighted average of local models. However, the aggregation process can inadvertently introduce performance-related bias when FL clients are assigned disproportionate weights. Clients with higher weights in the aggregation process have more influence on the global model than other clients, which may result in a model performance imbalance among clients, i.e., performance-related bias.

3.3.3 Evaluation Metrics of Performance-related Bias. Model quality metrics introduced in Section 3.2.3, such as accuracy, precision, recall, and F1-score, are also used to quantify performance-related bias in FL. In this context, these metrics evaluate performance disparities across FL clients, capturing the variations in the model's performance across different clients.

3.4 Contribution-related Bias

In the following, we first define contribution-related bias in FL, before presenting its causes, and the evaluation metrics to measure this type of bias.

3.4.1 Definition of Contribution-related Bias. Unlike other types of bias, which often focus on addressing disparities across different demographic groups or among different FL clients, the goal of mitigating contribution-related bias is to ensure that FL clients receive personalized models that reflect their individual level of contribution to the collaborative learning process. This type of bias refers to the misalignment between a client's contribution level and the performance of the model they receive. Let $Perf(\theta_i, D_i)$ denotes the model performance of the client C_i , and $L(C_i)$ the client's contribution level to the global model which can correspond to several factors such as data quality and volume, or the improvement in the global model's performance resulting from the client's local updates [147]. Formally, contribution-related bias occurs when the model performance does not align with the client's contribution level, as expressed below:

$$\exists i, j, \quad |Perf(\theta_i, D_i) - Perf(\theta_j, D_j)| < \epsilon \cdot |L(C_i) - L(C_j)|, \quad i \neq j,$$

where ϵ is a predefined threshold indicating acceptable alignment between performance disparities and contribution disparities. That is, high-contributing clients receive a better performing local model compared to their low-contributing counterparts. In practice, this means that clients actively contributing to the overall federated learning process through their larger and higher quality data should receive specialized models having higher performance, fostering an environment where meaningful contributions are acknowledged. This approach ensures that FL systems operate fairly, motivating participants to enhance model quality and overall system performance. Consider, for instance, larger banks that may be apprehensive about not receiving fair compensation and, as a result, choose not to collaborate with smaller banks. These smaller banks typically have a more restricted client base and, consequently, smaller contribution level. Rewarding the bank that contributes more with a model aligned to its level of involvement motivates active participation in the learning process.

3.4.2 Causes of Contribution-related Bias. Contribution-related bias in FL originates from several factors related to data distribution, client selection policies, and aggregation strategies. In the following, we focus on these key causes, highlighting how they specifically contribute to misalignment between client contributions and the quality of the model they receive.

Data Imbalance among FL Clients. Data imbalance among FL clients is a recurring FL issue driven by data heterogeneity, a defining characteristic of FL. This imbalance, previously discussed in the context of performance-related bias (see Section 3.3.2), is a key cause of contribution-related bias. Clients with imbalanced data contribute disproportionately to the training process, yet contribution-related bias arises when these clients receive the same global model, regardless of their individual contributions [126].

Bias Caused by FL Client Selection. Client selection policies, which also cause demographic and performance-related bias (see Sections 3.2.2 and 3.3.2), can introduce contribution-related bias in FL systems. When the server repeatedly selects a specific type of clients, these clients contribute more to the global model evolution while receiving a model whose quality does not align with their actual contribution levels. This misalignment becomes particularly evident when selection policies consistently select certain clients, allowing their contributions to dominate the training process. Meanwhile, clients making significant contributions, receive the same global model as all other clients, regardless of their varying levels of contribution. This results in an uneven distribution of benefits and reinforces contribution-related bias.

Bias Caused by FL Aggregation. Aggregation methods, which were earlier considered as demographic and performance-related biascauses (see Sections 3.2.2 and 3.3.2), can also be a source of contribution-related bias in FL. Typically, the FL global model is aggregated by applying a weighted average of clients' local models. However, these weights may not accurately represent each client's level of contribution. For example, some aggregation strategies assign higher weights to clients based on factors such as computational efficiency [143], which may not accurately represent their contribution levels. This misalignment leads to a discrepancy between the performance of clients' models and their true contributions.

3.4.3 Evaluation Metrics of Contribution-related Bias. Contribution-related bias is measured through client contribution and model quality metrics, that quantify the client's contribution level to the global model and the quality of the resulting FL model for each client, respectively. By evaluating the correlation between these metrics, contribution-related bias can be effectively assessed, revealing misalignments between client contributions and model performance. In the following, we provide a detailed overview of these metrics.

Client Contribution Metrics. Assessing client contributions in FL is important for tackling contribution-related bias. The metrics below outline various approaches to evaluate these contributions.

Reputation-based Client Contribution Evaluation relies on self-reported information, assuming clients can accurately and truthfully assess their circumstances, which is not always true in real-world FL scenarios. To address this, alternative approaches assess client contributions by analyzing model performance using reputation mechanisms [63, 164]. Some methods update reputation based on cosine similarity between a client's local model and the global model, while others use validation accuracy [90]. Additionally, there are further methodologies employing scoring functions [158], and mutual evaluation approaches [91].

Utility Game-based Evaluation is closely linked with profit-sharing frameworks, allocating rewards to FL clients based on their generated utility. In the marginal gain scheme, a client's payoff equals the utility gained by all clients when that participant joins. Conversely, the marginal loss scheme assigns a payoff based on the utility lost when a client leaves, highlighting the detriment of its departure. These schemes effectively assess clients' contributions, with the marginal loss scheme being the most widely used due to its practical significance [100].

Shapley Value-based Evaluation is a sophisticated and prevalent approach in FL [136, 139, 144], inspired by cooperative game theory [102]. It assigns a value to each client based on its marginal contribution to the learning process, quantifying each client's data and computational impact on overall model performance. With N FL clients and datasets $D_1, D_2, \dots, D_N$, a standard test set T , and constant C , the model trained on D_k is denoted by θ_k , and its performance on T is denoted by $U(\theta_k)$. The Shapley value ϕ_i quantifies individual client i 's contribution by averaging marginal contributions across all data subsets excluding i 's data, as outlined in Equation (11).

$$\phi_i = C \cdot \sum_{k \subseteq \{1, 2, \dots, N\} \setminus \{i\}} \frac{U(\theta_{k \cup \{i\}}) - U(\theta_k)}{\binom{N-1}{|k|}}. \quad (11)$$

Model Quality Metrics. Model quality metrics defined in Section 3.2.3 are involved in measuring contribution-related bias, where the correlation between clients' contribution levels and their models' quality reflects bias [90]. This correlation can be quantified using statistical methods, such as the Pearson correlation coefficient, to provide a measurable understanding of the bias [147].

4 Use Cases of Bias in Federated Learning

The use of FL has expanded across diverse fields, enabling collaborative model training while preserving data privacy. However, bias in FL systems can significantly impact their fairness, dependability, and overall performance, raising ethical and practical concerns, particularly in critical application domains. In order to show how bias arises and its potential consequences, this section examines actual instances of prejudice in FL in the fields of healthcare Section 4.1, autonomous vehicles Section 4.2, and financial fraud detection systems Section 4.3.

4.1 Bias in FL-based Healthcare Systems

FL is extensively used in the medical field since it allows several healthcare organizations to train prediction models cooperatively without compromising sensitive patient data privacy. It holds great potential for improving diagnostic accuracy, disease prediction, and treatment planning across diverse medical scenarios. However, during the federated training process bias can arise resulting in significant differences in model performance across demographic groups and healthcare organizations.

A concrete example of such bias can be observed in FL models used for cardiac arrest prediction. In this scenario, healthcare institutions collaborate to build a predictive model capable of estimating survival probabilities following cardiac arrest. However, when the training data is primarily derived from male patients, the resulting model tends to achieve significantly higher predictive accuracy for men compared to women. This bias results from the fact that men and women frequently have different cardiac arrest symptoms, risk factors, and physiological traits [71, 96]. When these gender-specific factors are underrepresented in training datasets, the resulting model may fail to generalize effectively for female patients, leading to inaccurate predictions and potentially suboptimal care recommendations. This disparity highlights how bias in FL-based healthcare systems can perpetuate and even exacerbate existing inequalities in medical care. Furthermore, beyond demographic bias, performance-related bias can also emerge in this context. For instance, larger hospitals smaller healthcare institutions with limited datasets might experience suboptimal performance from the shared model compared to larger hospitals with richer and higher-quality datasets, further deepening systemic disparities among FL clients.

4.2 Bias in FL-based Autonomous Vehicles

FL is being used more and more in autonomous vehicle systems to allow several platforms and manufacturers to train a global model without sharing sensitive data. This approach improves autonomous vehicle's collective intelligence, which helps them make better decisions in areas like navigation and pedestrian detection. However, bias may arise during the federated training process, leading to models that perform better for specific demographic groups. For instance, autonomous vehicles' pedestrian detection systems are a prominent example where bias can emerge. Research studies have shown that these systems frequently have worse detection accuracy for individuals with darker skin tones and for children. An analysis of eight AI-powered pedestrian identification models revealed that the detection accuracy of adult people was more than 20% higher than children, and it was over 3% higher for light-skinned than dark-skinned people [80]. The reason for this disparity is that darker-skinned people and children are often underrepresented in the datasets used to train these models, which results in models that are less effective at identifying these groups. This bias raises serious safety concerns since it may make it more difficult for autonomous vehicles to detect and react to certain pedestrians, increasing the likelihood of accidents involving these groups.

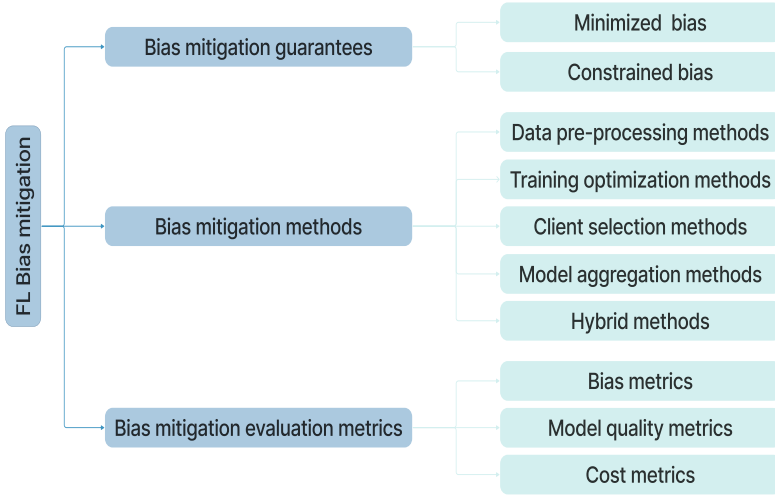


Fig. 2. Taxonomy of bias mitigation in FL.

4.3 Bias in FL-based Financial Fraud Detection

In order to jointly train fraud detection models across banks and financial platforms while protecting the privacy of sensitive consumer data, financial institutions are increasingly using FL [4]. These models aim at identifying fraudulent transactions in real-time and minimize financial losses. The federated training process, however, may give rise to contribution-related bias, resulting in misalignment between client contribution levels and the benefits derived from the global model. In a federated fraud detection system, large multinational banks and smaller regional banks collaborate to develop a common fraud detection model. Larger banks typically have access to extensive datasets with millions of transactions, including both fraudulent and legitimate patterns. In contrast, smaller banks often operate with limited datasets, characterized by lower transaction volumes and less diverse fraud patterns due to their smaller customer bases and regional focus. As a result, larger banks contribute significantly more to the global FL model. Despite their significant contributions, larger banks receive the same global model as smaller banks. This imbalance creates a misalignment between contribution levels and the benefits derived from the global model, as banks do not experience performance increases commensurate with their unique data contributions. Over time, these institutions may perceive the collaboration as unfair and decide to leave the federation process.

5 Federated Learning Bias Mitigation: A Taxonomy

To address bias concerns in FL, several approaches have been proposed. In Figure 2, we propose a taxonomy of bias mitigation in FL, and provide its overview with regard to the following aspects. (i) Different bias mitigation guarantees can be considered. We distinguish systems that aim at minimizing bias to the greatest extent possible, from those that maintain bias below a predefined threshold Section 5.1. (ii) There are different methods for FL bias mitigation, such as data pre-processing and model training optimization at the client-side, methods that are based on FL client selection and client model aggregation at the server-side, and hybrid methods Section 5.2. (iii) Additionally, we delve into the evaluation metrics used to assess the effectiveness of the proposed solution, including bias metrics, model quality metrics, and cost metrics Section 5.3. Each of these aspects is detailed in the following.

5.1 Bias Mitigation Guarantees

FL bias mitigation techniques fall into two main categories, each providing different bias mitigation guarantees to address the specific needs of the system. The first category encompasses methods that handle bias with the objective of minimizing it [91, 97]. On the other hand, other methods acknowledge the need to balance bias reduction with the specific FL application use cases [26, 33]. For instance, in many real-world scenarios, actions can be employed to ensure that some groups are not unfairly disadvantaged, often through the use of quotas. A prominent example is found in the United States, where a hiring rule referred to as the 80% rule or the four-fifths rule has been instituted to address hiring discrimination [55]. This rule mandates that the hiring rate of the unprivileged group should not exceed 80% of the hiring rate of the privileged group. In such scenarios, instead of striving for perfect parity between groups, these methods focus on maintaining bias below a predefined threshold that can be adjusted to align with the practical considerations of the considered legislation, enabling a flexible approach that tailors bias mitigation to meet the desired outcomes and fairness goals of the FL system.

5.2 Bias Mitigation Methods

Bias mitigation in FL can be applied at either the client or the server side, or a combination of both. When it comes to client-side FL bias mitigation, there are data pre-processing methods that aim at reducing or eliminate bias induced by the variation in data distributions. It focuses on the data representation before it is used for the training. It tackles bias within the client's local data by employing various algorithms or pre-processing steps such as data augmentation [34, 44]. Local training optimization methods are also applied at the client side during the training. These methods modify the training algorithms or loss functions to explicitly account for bias. They focus on mitigating bias locally at each FL client by incorporating bias constraints or regularization terms to penalize biased predictions [97, 107, 155]. There are also other techniques to address the potential bias such as model personalization methods and adversarial debiasing [46, 76, 155]. By doing so, the FL bias can be reduced on a per-client basis to ensure that each participating client contributes to an unbiased global model, thus, enabling fair decision-making and enhancing the overall effectiveness of the FL process.

Applying local debiasing at client level, relying on data pre-processing and learning optimization, may not provide global bias mitigation, given the usual data distribution heterogeneity (non-IID) among FL clients. To address this issue, several works follow a server-side approach, including methods employed during the FL client selection process [61, 160], or during client model aggregation [26, 79, 90, 108, 147, 150]. These methods aim at addressing bias on a global scale by carefully choosing participating clients or by adjusting the model aggregation process to mitigate bias. This adjustment involves reweighing local models, using gradient-based optimization methods, reputation mechanisms, or clustering techniques. Moreover, there are hybrid approaches that combine both server-side and client-side methods [38, 54, 58]. These approaches aim at leveraging the benefits of local bias mitigation at the client side while still achieving global bias mitigation at the server side [14]. In the broader context of ML bias mitigation, existing methods are usually grouped into three categories, namely *pre-processing*, *in-processing*, and *post-processing* [13, 47, 94]. In the specific case of FL systems, we categorize FL bias mitigation methods into: data pre-processing that also corresponds to its eponym in ML, local training optimization that align with ML in-processing, model aggregation that can be viewed as post-processing, and finally, client selection category that is specific to FL and does not directly fit into any ML category.

5.3 Bias Mitigation Evaluation Metrics

Evaluating the effectiveness of bias mitigation methods in FL involves using a set of metrics that capture different aspects of system performance, namely bias, model quality, and cost. The cost metrics are introduced below, whereas bias metrics and model quality metrics were previously introduced in Sections 3.2.3–3.4.3.

5.3.1 Cost Metrics. In order to assess the practicality of bias mitigation systems, it is important to evaluate their costs. Examples of cost metrics are provided below.

Computational Overhead is a pivotal cost metric in the evaluation of a FL bias mitigation system. Such an overhead may be due to additional computations performed for bias mitigation at the client side, for instance, for data pre-processing, or costly local model learning [107, 113]. Computational overhead can also be induced at the server side [53], due to costly approaches to mitigate bias during client selection, or during client model aggregation.

Communication Cost measures the amount of data exchanged over the network between the server and the clients, for FL bias mitigation [32, 34, 113]. The efficiency of a bias mitigation system in terms of data exchange contributes to minimizing communication delays, thus, ultimately enhancing the overall performance of the unbiased FL system.

Minimum Number of the FL Rounds Required to Mitigate Bias is considered by FL providers as a significant metric for mitigating bias [5, 150]. Resources and client availability limit the maximum number of rounds conducted simultaneously. Therefore, methods requiring fewer rounds offer greater flexibility and provide several advantages such as reduced communication bandwidth, lower energy consumption, and decreased communication delays.

6 Bias Mitigation Systems in Federated Learning

In the following, we explore existing FL bias mitigation systems outlined in Table 2, and highlight their characteristics. We categorize existing FL bias mitigation systems, first, by the type of FL bias they address, and then, by the technical methods they use to handle bias.

6.1 Systems Mitigating Demographic Bias

6.1.1 Bias Mitigation through Data Pre-processing. FL data pre-processing methods primarily focus on addressing bias arising from discrepancies in data distributions at the client side, such as data augmentation techniques. This approach enhances data diversity and minimizes the effects of distributional discrepancies to handle FL bias.

Data Augmentation. Data augmentation methods are used to mitigate demographic bias. For instance, FFL-OppoGAN generates synthetic data with opposite features to the original data to address both performance-related bias and demographic bias [42]. This augmented data is used in the FL process to decrease disparities in model performance among the FL clients and to guarantee fair treatment across the different demographic groups.

6.1.2 Bias Mitigation through Training Optimization. Training optimization methods tackle FL bias by explicitly incorporating bias constraints into the learning objective to effectively reduce bias. This category encompasses several methods, including regularization methods, and adversarial debiasing.

Regularization Methods. To mitigate bias in FL, regularization methods include bias constraints in the loss function by penalizing models that show prediction disparities. Several exiting works apply such a method [33, 138, 155]. For instance, in AgnosticFair [33], an extra term is added

Table 2. Overview of FL Bias Mitigation Systems for Different Types of Bias, Bias Mitigation Methods (Data preprocessing - Data., Local Training Optimization - LTO., Client Selection - SeleC., Model Aggregation - Agg.), Bias Mitigation Guarantees, Evaluation Metrics

FL bias type	Bias mitigation system	Bias mitigation method				Bias mitigation guarantees		Evaluation metrics				Sensitive attributes in demographic bias mitigation	
		Data.	LTO.	SeleC.	Agg.	Minimized bias	Constrained bias	Demographic bias	Model quality	Client contribution	Cost	Single attribute	Multiple attributes
Demographic bias	FFL-OppoGAN [42]*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	AgnosticFair [33]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FADE [46]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FPFL [103]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FairVFL [113]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FPFR [138]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	GIFAIR-FL [155]*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	mFairFL [128]*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FedUFO [161]*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FairFL [160]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	F3 [61]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FedHeteroFair [60]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FCFL [28]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FedMinMax [108]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	ASTRAL [150]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FairFed [38]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	MLPFL [24]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FairTrade [5]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	FairSCAT [149]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Performance-related bias	Fed-ZDAC [44]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	ASTRAEA [34]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Ditto [76]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedMDFG [107]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	AFL [97]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CoreFed [116]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedNLR [137]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	SRA [83]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	RBCS-F [53]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedHD [87]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FairJAW-FL [167]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedSDR [92]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	E3CS [52]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedFV [141]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedMGDA+ [50]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	q-FFL [79]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Fed-ZDAS [44]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedHEAL [20]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedLF [106]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedMC [123]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedCE [58]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Fedra [54]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Prop-FFL [49]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Fjord [48]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Ada-FFL [25]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedGC [23]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedUB [163]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedFTP [153]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FairDPFL-SCS [118]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CosPer [117]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	PPDHFL [72]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
Contribution-related bias	FPPDL [91]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	IFL3A [70]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CA [151]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CareFL [88]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	RNS [89]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CFLL [90]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	CGSV fair reward [147]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	ConFedIn [62]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	πFL [19]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedAVE [142]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FedCom [162]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Fedcav [156]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	FAIR-CLAD [135]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	Fair-UDPFL [2]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	F-GRACE [105]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A
	HFedCWA [32]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	N/A	N/A

*indicates systems mitigating both demographic and performance-related bias.

to the loss function to enforce demographic bias mitigation, resulting in fairer models. GIFAIR-FL penalizes variability in clients' losses and demographic groups [155], guiding the optimizer toward fair solutions, addressing both demographic bias and performance-related bias.

Adversarial Debiasing. Adversarial debiasing mitigates demographic bias by incorporating an adversary that tries to predict sensitive attributes from the model's outputs. During training, the model is optimized to perform well on the main task while minimizing the adversary's ability to detect sensitive attributes, thus reducing the model's reliance on these attributes. This approach helps create a more fair and unbiased model. For instance, FADE allows FL clients to train the model and adversarial component with local data [46]. Each client's model includes a decoder, encoder, and group discriminator. The goal is to minimize task and adversarial losses through

a min-max problem, achieving adversarial balance when the discriminator cannot differentiate between privileged and unprivileged groups. FairVFL enhances VFL model fairness by learning unified, fair sample representations [113]. Clients first learn local data representations, which are then aggregated at the server. This unified representation, preserving data privacy, can encode bias toward fairness-sensitive attributes. To mitigate this, FairVFL uses adversarial learning to remove bias from these representations, ensuring fairness while maintaining privacy.

6.1.3 Bias Mitigation through Client Selection. Client selection-based bias mitigation in FL strategically selects clients to address bias. For instance, in FairFL [160], a unique client selection method optimizes bias function approximation and model accuracy using clients' collected measurements. It introduces a team Markov game model for collective decision-making in the local update process, overcoming FL's black-box nature. F3 and FedHeteroFair both address demographic bias, proposing several client selection heuristics. F3 focuses on face attribute classification by balancing fairness and accuracy with three heuristics [61]: FairBest selects the model that minimizes fairness violations on a public dataset available at the server side; α -FairAvg generalizes FairBest by selecting the top $\alpha\%$ of local models; and α -FairAccAvg selects the top $\alpha\%$ of local model parameters with the optimal accuracy-to-fairness violation ratio. Similarly, FedHeteroFair [60], which stands for fair FL for heterogeneous data, mitigates demographic bias despite data heterogeneity. In addition to techniques similar to F3's, it introduces FairAccDiff, which selects clients by maximizing the value of twice the accuracy minus the fairness loss.

6.1.4 Bias Mitigation through Model Aggregation. Aggregation-based bias mitigation methods are designed to minimize bias at the global model update process. These strategies acknowledge the possible presence of biased data and models at the individual client level, which can impact the overall model. To counteract this, several approaches have been proposed. Among them are model reweighing methods, which alleviate bias by assigning varying weights to the local models during the aggregation process to derive a fair global model. Gradient-based methods are also prevalent, seeking to minimize the bias of the global model by optimizing the client gradients.

Local Model Reweighing. Local model reweighing methods tackle bias by adjusting the aggregation strategy and assigning varied weights to FL clients' local models. A notable example is ASTRAL [150], which reweighs local models based on their impact on global model bias and accuracy, aiming at mitigating demographic bias while preserving model accuracy. Furthermore, ASTRAL handles bias with regard to multiple sensitive attributes, through a multi-objective optimization approach that aims at maximizing model accuracy while minimizing the bias related to each sensitive attribute. Similarly, FedMinMax updates weighting coefficients using a projected gradient ascent step based on clients' shared parameters and bias values [108].

Gradient Optimization-based Methods. Gradient optimization-based methods address bias in the FL process by adjusting model parameters based on the gradients of the loss function. These methods enable nuanced updates that prioritize bias reduction and promotes fair model convergence. One example is FCFL [26], which constrains demographic bias below a fixed threshold while maintaining consistent model performance. It formulates a multi-objective constrained optimization problem to enforce the global model to respect the defined bias constraint on each client's distribution without compromising accuracy, achieving Pareto optimality.

6.1.5 Hybrid Methods for Bias Mitigation. Hybrid methods combine various approaches for bias mitigation [5, 24, 38, 128, 149, 161]. For instance, FairFed combines local debiasing with a fairness-aware aggregation method, adjusting clients' model weights based on discrepancies between local and global fairness, favoring clients aligning with the overall demographic fairness [38]. MLPFL

uses a multi-layer personalized FL framework based, on the one hand, on a global model which is trained in a distributed fashion via meta-gradient updates, and on the other hand, on personalized models for different demographic groups for a fair model over the different groups [24]. FairSCAT uses semi-centralized adversarial training with a **Variational AutoEncoder (VAE)** to generate adversarial samples, with the encoder on the client side to encode local samples into feature dimensions, which ensures the privacy of client data [149]. Symmetrically, the decoder that generates adversarial samples is at the server side to improve the demographic fairness of the model. In [161], the authors consider FL-based healthcare systems, which may lead to an unfair model that performs poorly with underrepresented demographic groups. They propose FedUFO, which leverages VAE and robust optimization for balanced utility and fairness.

6.2 Systems Mitigating Performance-related Bias

6.2.1 Bias Mitigation through Data Pre-processing. To address performance-related bias, data pre-processing methods leverage data augmentation techniques to enhance data diversity and minimize the impact of distributional discrepancies among FL clients, thereby mitigating bias. These methods generate synthetic training samples to expand datasets, allowing FL models to better adjust to data distribution variations without additional data collection. These techniques are particularly valuable to mitigate performance-related bias in FL setup. For instance, Fed-ZDAC employs zero-shot data augmentation to promote consistent accuracy performance across FL clients by introducing synthetic data to supplement the FL training with underrepresented samples [44]. Similarly, Astraea addresses imbalances through Z-score-based data augmentation and a mediator-based multi-client rescheduling strategy [34], mitigating global data imbalance by applying zero-shot data augmentation, thereby fostering more uniform accuracy performance among FL clients.

6.2.2 Bias Mitigation through Training Optimization. To tackle performance-related bias through training optimization methods, we identify either regularization or personalization approaches that effectively mitigate bias.

Regularization Methods. Several bias mitigation systems rely on regularization methods to ensure fairness across FL clients. One example is SRA that adds a bias term to ensure uniform performance across all FL clients [83]. The fairness term is designed to reduce the variation in performance among FL clients, ensuring that clients with poor communication links or low computational resources are not unfairly disadvantaged. While, FedMDFG employs a fair driven objective to guide gradient updates [107], minimizing the angle between client models' performance vectors and a fair guidance vector, promoting uniform model performance among FL clients.

Personalization Methods. Personalization methodologies involve developing tailored models that adjust to the unique characteristics and requirements of individual FL clients, addressing FL bias by customizing the learning process to enhance fairness. One example is GIFAIR-FL-Per [155], a personalized fair algorithm where GIFAIR-FL is slightly tailored by adding a client-specific term to the loss function, introducing a device-specific solution with minimal discrepancies among all FL clients. Likewise, Ditto [76], a personalized multitask FL algorithm, addresses performance-related bias. It treats the global and local learning objectives as distinct tasks, aligning personalized models with the optimal global model to promote fairness among FL clients. Another approach is FedNLR, which uses neuron-wise learning rates during the local training process [137]. This method enhances the learning of neurons bound to local classes on local data, while minimizing the decay of non-local class knowledge, reducing the neurons activation on samples from classes not present in the client's data. This ensures that neurons specialized for local classes are effectively trained while maintaining the global model's performance across all classes.

6.2.3 Bias Mitigation through Client Selection. Two additional works address performance-related bias caused by repeatedly selecting the same clients in each FL round. RBCS-F ensures fair and balanced client selection using Lyapunov optimization and a contextual combinatorial multi-arm bandit model [53], considering clients contexts and historical performance to ensure that all clients, regardless of their performance, have a fair chance of being selected for training. FedHD [87] uses stratified sampling combined to multi-arm bandit techniques to mitigate heterogeneity and reduce bias, exploring client combinations for a global data distribution match. Similarly, E3CS optimizes client selection probability to maximize the model's worst performance across any training data mixture [52]. FairUAV-FL [167] selects clients balancing model freshness and energy consumption to improve generalization and reduce performance disparities among FL clients. Furthermore, FedSDR [92] selects the most representative clients based on data distribution to optimize the global model.

6.2.4 Bias Mitigation through Model Aggregation. Several aggregation-based methods have been proposed to mitigate performance-related bias, including local model reweighing, data-based, and gradient optimization approaches, each aiming at improving fairness and enhance the global model's performance among FL clients.

Local Model Reweighing. To address performance-related bias, q-FFL reweighs aggregated loss [79], assigning higher weights to poorly performing devices to ensure a uniform accuracy distribution. FedHEAL adjusts the aggregation weights based on the distances between global and local models, ensuring fairness among FL clients [20]. While also considering the consistency of parameter updates, retaining only significant updates. Similarly, FedCE introduces new weighting factors derived from the differences in gradient directions and the global model prediction errors calculated by excluding the client's own parameters from the global model [58]. It gives higher weights to clients with unique gradient information or high prediction errors, reducing performance disparities among clients.

Data-based Methods. An alternative strategy for mitigating bias is through server-side training, wherein the server generates an equivalent quantity of synthetic data from each client model received, consolidating all synthetic client data into a well-balanced synthetic dataset. The server aggregates all received client models into a global model, and then trains this model on the synthesized dataset. This approach is illustrated by the methodology adopted in Fed-ZDAS to mitigate performance-related bias [44].

Gradient Optimization-based Methods. These gradient optimization-based methods mitigate performance-related bias by resolving gradient conflicts among FL clients, thereby reducing disparities in their model performance. Among these methods, FedFV addresses performance-related bias using cosine similarity to detect and resolve gradient conflicts [141]. This method adjusts gradient direction and magnitude to manage client conflicts, aiming for Pareto stationary solutions and enhancing fairness among FL clients. Furthermore, FedMGDA+ also mitigates performance-related bias through multi-objective optimization [50]. It optimizes the loss function for each client individually and simultaneously, seeking a shared descent direction that benefits all clients without compromising model performance. FedLF formulates a multi-objective optimization problem with a fair-driven objective to mitigate gradient conflicts at both model and layer levels [106], using a layer-wise multiple gradient descent algorithm. Moreover, FedMC refines the global gradient by introducing a sub-optimization objective to alleviate conflicts between the global and local gradients with the largest deviation [123], improving the performance and convergence rate of clients with initially poor performance, thus reducing bias.

6.2.5 Hybrid Methods for Bias Mitigation. There are several systems that follow a hybrid approach for mitigating FL performance-related bias [23, 25, 48, 49, 54, 117, 118, 153, 163]. For instance, Prop-FFL applies regularization and maximizes the product of relative local models losses during aggregation, ensuring uniform performance across FL clients [49]. FjORD employs ordered dropout and knowledge distillation to tailor model training to client capabilities and ensure uniform model performance distribution among FL clients [48]. Similarly, PPDHFL uses ordered partitioning and a game-theoretic approach to reduce client performance disparities [72]. FedFa uses a federated optimization algorithm with double momentum gradients and a participation-based weighting strategy [54].

6.3 Systems Mitigating Contribution-related Bias

In the following, we present various systems designed to mitigate contribution-related bias, highlighting their key approaches and the methods they use to address this issue effectively.

6.3.1 Bias Mitigation through Training Optimization. To mitigate contribution-related bias, training optimization methods mainly rely on personalization approaches. One such example is FPPDL [91], which uses a local credibility mutual evaluation mechanism to mitigate contribution-related bias through personalization. Clients earn points by contributing information, then these points are used to trade information with others, encouraging the upload of more samples or gradients to earn points and then use them to download more gradients from the others.

6.3.2 Bias Mitigation through Client Selection. To mitigate contribution-related bias, several works use client selection strategies to ensure fairness. RNS employs an optimal multidimensional contract [89]. This contract is designed to guarantee reliability, honesty, and fairness, thus maximizing learning accuracy. It incorporates rules to ensure that the rewards for FL clients are fair based on their contributions. Similarly, CA selects local models based on their quality and the client computational capabilities [151]. Through a contract theory, only the best models are chosen for aggregation. This selective process ensures that the global model is updated with the highest-quality contributions, aligning client contribution levels with model performance. Moreover, IFL3A [70], which refers to Incentive FL mechanism in wireless cellular networks: An Auction Approach, formulates client selection as an auction game. It uses a primal-dual greedy auction mechanism where each client optimally allocates resources and local accuracy. Then, the server determines the participants based on potential contributions. Another approach, CAreFL [88], focusing on bias mitigation in smart healthcare, uses a GTG-Shapley approach to provide fair and accurate client contribution values. Based on these values, the server selects the optimal subset of participants whose local model updates yield the best-performing aggregated FL model.

6.3.3 Bias Mitigation through Model Aggregation. Aggregation-based methods are employed to address contribution-related bias in FL. These methods include several approaches, such as local model reweighing, reputation mechanisms-based, gradient optimization-based, and clustering-based methods. In the following, we will introduce each of these approaches.

Local Model Reweighing. Local Model Reweighing methods address contribution-related bias by adjusting local model weights based on clients contributions. For instance, both FedCom and FedCav use a contribution-aware aggregation approach to effectively mitigate bias [156, 162]. Furthermore, π FL mitigates bias assigning aggregation weights based on FL clients' contribution while introducing a multi-party computation-based shapley value for enhanced privacy [19].

Reputation Mechanisms-based Methods. Reputation mechanisms-based methods leverage the concept of clients reputation to promote fairness and alleviate bias in FL systems. The

core principle involves assigning reputation scores to FL clients that are allowed to download aggregated updates according to their reputation levels. These methods are mainly used to tackle contribution-related bias. For instance, both Collaborative Fair FL shorten CFFL, and FedAVE use a reputation mechanism to enforce the FL clients to converge to different models [90, 142], the reputation values are maintained by the FL server which evaluates the clients contributions during the learning process and iteratively updates their respective reputations. The FL clients are allowed to download updates according to their reputation scores, achieving fairness without compromising the predictive performance.

Gradient Optimization-based Methods. Gradient Optimization-Based Methods focus on adjusting the model's gradient updates to mitigate contribution-related bias and ensure fairness in the aggregation process. For instance, a cosine gradient Shapley value is introduced in [147], aiming at impartially assessing each FL client's expected marginal contribution. Based on this metric, a training-time gradient reward mechanism with a fairness guarantee is proposed, using a sparsifying gradient trick to control the quality of the aggregated parameters. Consequently, clients contributing higher-quality parameters are rewarded fairly.

Clustering-based Methods. FL clustering-based bias mitigation methods primarily address contribution-related bias by grouping FL clients based on their respective contributions. In a given cluster, clients are expected to have similar contribution profiles, leading to the convergence on a shared final model. This ensures that the model is updated in alignment with the clients' contribution significance. For instance, ConFedIn [62], based on contract theory for FL incentives, addresses contribution-related bias by clustering clients based on data quality and computational resources. Each cluster receives a tailored contract, with higher quality clients earning more rewards. This ensures fair compensation for FL clients. Similarly, FAIR-CLAD [135] uses contribution scores to assess client contributions and clusters clients accordingly, assigning each cluster a distinct global model for fairer updates.

6.3.4 Hybrid Methods for Bias Mitigation. Hybrid methods for contribution-related bias mitigation combine local training optimization and model aggregation-based methods. For instance, Fair-UDPFL controls the quality of the global model shared with FL devices based on their contributions, using differential privacy to limit model divulgence for each device according to its contribution [2]. It also balances model quality with adaptive computation, adjusting each device's training efforts, such as the number of local epochs. HFedCWA [32] combines weighted aggregation, assigning higher weights to devices with greater contributions, and personalization for optimal device engagement. Similarly, F-GRACE mitigates bias through local training optimization via gradient alignment and reward, and model aggregation using motif prototypes to capture key data patterns [105]. This ensures the global model maintains accuracy while promoting fairness across the different FL clients.

6.4 Applications and Evaluations of Bias Mitigation Systems

Applications of bias mitigation methods in FL span a wide range of domains, such as healthcare [26, 46, 150], finance [33, 46], and social media [76, 97]. Table 3 provides an overview of these application domains, highlighting the datasets and models used to evaluate the effectiveness of the existing methods under each domain, along with the theoretical guarantees supporting their results. To evaluate these methods, comprehensive benchmarks have been developed to assess their performance in practical settings, using diverse datasets including tabular, image, text, and graph formats. Particularly, demographic bias mitigation methods, are tested on datasets containing sensitive attributes such as gender, age, and race in the Adult dataset, and gender and age in CelebA [150].

Table 3. Applications and Evaluations of FL Bias Mitigation Systems

Applications	References	Datasets	Models	Theoretical guarantees
Healthcare and Human Activity Recognition	[26, 38, 46, 49, 58, 72, 105, 150, 153, 161]	eICU, MEPS, Prostate MRI dataset, Retinal fundus dataset, Histopathology-Kidney, Histopathology-Lung, Fetal, Prostate, COVID-19, Support, Fitzpatrick, MCI, PROTEINS, DD, MobiAct, ARS, TILES, UCF101, HAR	CNN [46, 49, 72, 153], FCN [150, 161], Linear models [26, 38, 150], RNN [46], GNN [105], Transformer models [153]	Bias-constrained convergence [46, 49, 161], Multi-objective optimality in client selection and model aggregation [26, 58, 72]
Finance	[5, 26, 33, 38, 46, 50, 90, 97, 103, 108, 116, 128, 150, 160]	Adult, Bank, Default, ACS Employment (ACSIIncome), Dutch, KDD	CNN [46], FCN [5, 90, 103, 108, 113], Linear models [26, 33, 38, 50, 97, 108, 116, 150, 160], RNN [46], AutoInt [113], ReLU-NN [128]	Bias-constrained convergence [33, 46, 97, 116, 128], Multi-objective optimality in client selection and model aggregation [26, 50, 108, 116, 128]
Education	[5, 24, 48, 50, 54, 79, 155]	Shakespeare, ASSISTments, EdNet, FOC, QDS, OSFETs, Law	FCN [5], RNN [48, 50, 54, 79, 155]	Bias-constrained convergence [79, 155], Multi-objective optimality in client selection and model aggregation [48, 50]
Crime and Justice	[38, 128, 160]	Crime, COMPAS	Linear models [38, 160], ReLU-NN [128]	Bias-constrained convergence [128], Multi-objective optimality in client selection and model aggregation [128]
Transportation	[25, 76, 79, 151]	Vehicle, BelgiumTSC, Waymo Open Dataset, KITTI Dataset	CNN [151], Linear models [25, 76, 79]	Bias-constrained convergence [76, 79]
Entertainment and Recommendation	[97, 113, 138, 147]	Movie review, Cornell Movie, Yelp, IMDB, Last.fm, NEWS, Filmtrust, Steam-200k, MovieLens-100K, MovieLens-1M, ADS	CNN [147], RNN [97], GNN [138], Recommendation models (NAML [113], LSTUR [113], NRMS [113]),	Bias-constrained convergence [97, 147]
Social Media and Communication	[25, 54, 76, 79, 97, 105, 147, 153]	Sent140, CrisisMMD, Stanford Sentiment Treebank, IMDB-BINARY, StackOverflow, X (formerly known as Twitter) Sentiment Analysis Dataset, Penn TreeBank	CNN [147, 153], Linear models [76], RNN [25, 54, 79, 97], Transformer models [153], GNN [105]	Bias-constrained convergence [76, 79, 147]
Computer Vision	[2, 19, 20, 23, 32, 34, 44, 44, 48–50, 52–54, 61, 62, 72, 76, 79, 87–92, 97, 106–108, 113, 116–118, 123, 135, 137, 141, 142, 147, 150, 151, 153, 155, 156, 162, 163, 167]	FMNIST, EMNIST, FashionMNIST, FEMNIST, CIFAR-10, CIFAR-100, Tiny-ImageNet, SVHN, Omniglot, DIGITS, DomainNet, PACS, CINIC-10, Digit-Five, Digits, USPS, SYN, Office-Caltech, Caltech, Amazon, Webcam, DSLR, Clothing1M, CelebA, UTKFace, FairFace, FFHQ	CNN [2, 19, 20, 32, 44, 48–50, 52, 53, 61, 72, 76, 79, 89, 91, 92, 106–108, 113, 116–118, 123, 135, 137, 141, 142, 147, 150, 151, 153, 155, 156, 162, 163], FCN [19, 89, 90, 106, 107, 163], Linear models [54, 88, 97], SecureBoost Tree [88], FNN [141], FedGC model [23]	Bias-constrained convergence [34, 49, 76, 79, 97, 107, 116, 137, 147, 155, 156], Multi-objective optimality in client selection and model aggregation [48, 50, 52, 53, 62, 72, 89, 106–108, 116, 123, 141, 151, 162]

Furthermore, depending on the domain and data-specific needs, the mitigation methods are evaluated on different learning models ranging from traditional linear models [26, 38] to advanced architectures such as **fully connected networks** (FCN) [103, 161], **Convolutional Neural Networks** (CNN) [46, 49, 72, 153], **Recurrent Neural Networks** (RNN) [25, 54, 79, 97], **Graph Neural Networks** (GNN) [138], and transformers [153]. The results demonstrate the ability of FL bias mitigation methods to address bias effectively across various models, even with complex architectures like GNNs and transformers. The evaluation further emphasizes the system's robustness in realistic FL scenarios, particularly to data heterogeneity, ensuring that bias mitigation methods perform reliably under non-IID data distributions [38, 46, 49, 58, 61, 150, 153, 161], and scalability, assessing the effectiveness of existing methods as the training dataset size [90, 150] or the number of participating FL clients changes [5, 33, 46, 90, 116, 150]. This ensures the applicability of these mitigation methods in diverse FL deployments, providing critical insights into their practical reliability.

To further support the effectiveness of these methods, some provide a theoretical analysis, including bias-constrained convergence and multi-objective optimality in client selection and model aggregation. Bias-constrained convergence analysis mainly demonstrates that the learning model converges to a fair solution. For example, regularization-based methods that ensure fairness by incorporating bias-related terms into the loss function show that their proposed approach converges to a fair model with the modified loss [33, 46, 97]. Other methods, particularly those focused on mitigating performance-related bias, demonstrate alignment of local client models with minimal variation in their performance, leading to improved uniformity across FL clients [49, 79]. On the other hand, client selection and model aggregation optimality approaches guarantee the fairness of the global model by demonstrating the effectiveness of their solutions. For instance, client selection-based methods define selection regret functions and show their convergence to near-optimality [52, 53]. Other aggregation-based methods prove Pareto optimality, ensuring that no client's performance can be improved without negatively affecting another's [50, 107, 108, 116, 141]. Moreover, contribution-related bias mitigation methods offer theoretical guarantees through optimal client selection and model aggregation, leveraging smart contracts and auction mechanisms to ensure that FL clients are rewarded according to their contribution levels [62, 89, 147, 151, 162]. Furthermore, some methods analyze both bias-constrained convergence and client selection and model aggregation optimality. For instance, in mFairFL, the framework is proven to converge to an optimal solution for demographic fairness by introducing a demographic bias term in the loss function. Additionally, the method theoretically shows how it resolves conflicts among gradients from different clients, minimizing variance in model performance to mitigate performance-related bias [128].

6.5 Summary

To promote fairness and counteract FL bias, a variety of methods have emerged. Most existing methods strive to minimize bias as much as possible, while only a few focus on maintaining it below a given threshold. Furthermore, the challenge of handling bias stemming from multiple sensitive attributes is tackled by some existing works [113, 150, 155, 160, 161]. Overall, bias mitigation methods include data pre-processing techniques, aiming at handling demographic bias related to a single sensitive attribute, or to handle performance-related bias [34, 44]. There are also hybrid methods that combine different techniques to mitigate FL bias [48, 54, 72].

Different application fields were considered in existing works on FL bias mitigation, including healthcare, finance, transportation, and so on. To evaluate the proposed bias mitigation methods, several metrics have been proposed, including SPD and EOD for demographic bias [26, 38, 150], model accuracy for performance-related bias [50, 76, 97], and the Pearson coefficient for contribution-related bias [90, 147]. These metrics are interdependent, and their interactions need further investigation. Additionally, aside from bias metrics, other measures assess the cost of the bias mitigation methods and their impact on model utility. Furthermore, some articles provide a theoretical analysis of their solutions, proving, for instance, the convergence of bias to its limit, or the optimality of the aggregated FL model.

In summary, in FL bias mitigation, the different types of bias (i.e., demographic Bias, performance-related bias, and contribution-related bias) can overlap, and the mitigation of one type having a side-effect on another type of bias. The interaction among these bias is largely unexplored, presenting a promising research avenue for developing comprehensive bias mitigation strategies. Furthermore, the architecture of FL systems can also have an impact on bias mitigation. These challenges are further discussed in the following section.

7 Challenges and Open Research Directions

Although several FL bias mitigation systems exist, many research directions remain open. These unexplored avenues are crucial for addressing the ongoing challenges in mitigating FL bias. They include the automatic detection of other attributes inducing demographic bias, bias mitigation in vertical FL frameworks, exploring bias metrics interdependence, examining bias-robustness-privacy tradeoffs, and developing bias mitigation approaches for multimodal FL scenarios remain under-explored. In this section, we present the key open research directions in the field.

7.1 Identifying Attributes Indirectly Causing Bias

While classical demographic bias mitigation approaches focus on legally defined sensitive attributes, such as race and gender, to delineate demographic groups [35, 145], emerging research reveals that bias persists even when these attributes are excluded from the dataset and not considered during the learning process [31, 110]. Several studies have shown that bias can arise from intricate correlations between sensitive attributes and other dataset attributes [166]. This highlights the critical need for innovative solutions that automatically identify the (non-sensitive) attributes that indirectly generate bias, and handle FL model fairness accordingly. This is inherently challenging due to the non-IID decentralized data distribution in FL systems, without a global knowledge of statistical properties of FL clients' data. Despite these challenges, researchers can explore innovative approaches that leverage available information and incorporate domain knowledge to detect attributes that cause bias.

7.2 Impact of Bias Mitigation on Model Quality

In FL, the analysis of bias metrics interdependence takes on increased significance, especially when considering the exploration of tradeoffs with model accuracy. This sophisticated area of research aims at exploring the complex interplay between various bias metrics in FL and their potential negative side-effect on model quality and accuracy. While this interplay has been extensively examined in traditional ML [1, 27, 65, 84], it remains relatively unexplored in FL, where bias can arise in complex patterns, influenced by unique data characteristics among the FL clients and the local model training. The distributed and decentralized nature of FL raises additional challenges in terms, both of identifying bias, and evaluating its cumulative impact on model quality. The development of methodologies that accurately interpret and assess the interconnections among different bias metrics as well as model quality is a pivotal key to achieve a harmonious balance, which involves reducing bias while preserving model quality in FL systems. The objective is to find an optimal equilibrium that enhances both fairness and FL predictive accuracy. Such a strategic approach is essential for the progression of FL applications, as it promises to deliver results that are not only more dependable, but also ethically sound.

7.3 Tackling Different Types of Bias

Recent studies have explored methods to mitigate bias across different demographic groups, ensure uniform performance among FL clients, and reward clients based on their contributions. While progress has been made in addressing these issues separately, it is crucial to consider them as a whole, as different types of bias are interlinked. For instance, demographic bias mitigation strategies might inadvertently exacerbate performance-related bias by giving significant weight to clients with demographically unbiased models. This increases their influence on the FL global model [38, 150], leading to greater performance disparities among clients. Some studies have tackled both demographic and performance-related bias simultaneously, achieving uniform performance across clients and demographic groups (e.g., [155]). However, these methods are mainly

effective when clients have data pertaining to a single demographic group. Furthermore, to our knowledge, no studies have addressed both demographic bias and contribution-related bias simultaneously. The convergence of these diverse bias types presents a complex challenge, as they may overlap, interact, and potentially amplify each other. Specifically, addressing demographic bias alongside either performance- or contribution-related bias yields valuable insights and remains as a significant area of ongoing research.

7.4 Bias Mitigation in Vertical Federated Learning

In real-world applications, data often includes multiple sensitive attributes, leading to bias. In VFL, where data is partitioned vertically, each client holds different features for the same entities. This causes VFL models to inherit demographic bias from various parties. Consequently, bias detection in VFL requires integrating features from multiple clients, making the detection and mitigation process more challenging due to limited data visibility [86]. Moreover, existing bias mitigation methods typically rely on storing sensitive attributes within each client's data [26, 38, 150], making them generally inapplicable to VFL scenarios. Additionally, evaluating the contribution of different clients to the learning process is also more complex in VFL, as it involves coordinating data with varying quality levels. This complexity is further compounded by the need to ensure data privacy and security while managing communication overhead.

Despite progress in FL bias mitigation, few studies focus specifically on VFL and are specifically tailored to handle demographic bias [108, 113]. These methods often face scalability issues. For instance, communication rounds in FairVFL can become bottlenecks in large-scale FL [113]. Furthermore, to the best of our knowledge, there are no works specifically tackling performance- and contribution-related bias in VFL. This gap highlights the need for further research and effective bias mitigation methods to enhance the fairness and efficiency of VFL systems, especially as they are increasingly used in fields such as IoT and finance [36, 73, 85].

7.5 Bias Mitigation vs. FL Privacy

FL paradigm comes with privacy guarantees preventing the clients from directly sharing their private data, reducing the risk of direct data exposure during training. However, it remains vulnerable to several privacy risks, such as membership inference attacks, where adversaries determine if specific data points were part of the training dataset, and model inversion attacks, which attempt to reconstruct the private client data [29, 41]. These attacks exploit the exchanged model updates and the iterative nature of FL, potentially leaking private information despite data locality guarantees.

To counter privacy attacks in FL, **differential privacy (DP)** is among the most widely used protocols, adding noise to model parameters to protect them from potential threats. However, applying DP to a biased model can further amplify existing disparities [6], the added noise disproportionately reduces the model performance for under-represented groups, thereby increasing bias. On the other hand, models trained with fairness constraints are more vulnerable to privacy attacks [16], particularly for under-represented groups. Bias mitigation algorithms often tend to focus more on memorizing data from under-represented groups to equalize error rates across demographic groups, inadvertently increasing information leakage about these groups. This reveals a critical privacy-fairness tradeoff, where improving one objective compromises the other. Future research must explore frameworks to balance these dimensions effectively.

7.6 Bias Mitigation vs. FL Robustness

In addition to bias concerns, FL presents a significant challenge when it comes to ensuring robustness. The presence of malicious parties, referred to as Byzantine clients, can threaten

the integrity of the FL process. These Byzantine clients deviate from the standard protocol and send arbitrary updates to the server. For instance, in a FL system where honest clients regularly send updates to the server, a Byzantine client can manipulate an update to make the server compute any desired global model. Addressing the presence of Byzantine clients and mitigating model bias are critical requirements in FL systems. Existing works focus on tackling these issues separately. Some research efforts are dedicated to developing Byzantine resilient FL methods [11, 111, 152], while others investigate methods to mitigate bias. However, combining these goals into a single FL system is challenging due to conflicting objectives. Bias mitigation requires considering outlier clients to enhance the representation of their local data, while robust FL systems may eliminate outlier clients, potentially worsening model bias. There are numerous research directions that remain unexplored. One interesting direction could be reducing the misidentification of Byzantine clients, as these clients likely possess valuable data representing underprivileged groups. Their participation in the FL process could help to bring down the global model bias. The server can attempt to identify such honest and minority clients among the presumed Byzantine clients by leveraging robust aggregators. To distinguish these clients from actual Byzantine ones, reputation mechanisms could be employed to accurately identify Byzantine clients.

7.7 Bias Mitigation in Multimodal FL

Multimodal FL has emerged as a recent paradigm, leveraging diverse data modalities from various devices. While the potential of multimodal FL is vast [30, 120, 146], a noteworthy gap in research lies in the absence of dedicated bias mitigation methods within this framework. The integration of multiple modalities gives rise to a complex interplay of bias, that requires focused attention. The exploration of how multimodality impacts bias and the formulation of effective bias mitigation strategies within this evolving framework represent an interesting open research avenue. Understanding the intricate dynamics of bias across diverse modalities and developing methods that account for these nuances is crucial for ensuring fairness and accuracy in multimodal FL. As this paradigm gains prominence, the imperative to address bias becomes increasingly apparent, marking a compelling frontier for researchers to explore and contribute innovative solutions to mitigate bias in multimodal FL.

8 Conclusion

In FL, bias presents a multifaceted challenge that significantly influences the effectiveness and fairness of the learning process. This survey, conducted through a systematic literature review, offers a comprehensive overview of FL bias. First, we define bias and categorize its diverse manifestations in FL into three types: demographic bias, focusing on disparities among the different groups defined by the sensitive attributes; performance-related bias, measuring disparities among FL client performance; and contribution-related bias, referring to the imbalance between a client's contribution and the model it receives. Subsequently, we present the root causes of this issue, followed by a taxonomy that contributes to a comprehensive understanding of bias mitigation methods. Further, the survey explores the existing bias mitigation systems employed to mitigate the different FL bias types. This review provides a thorough discussion on FL bias and current methods to address it, highlighting challenges that are still not well addressed in the literature, concluding that several questions still require investigation. Thus, the present work contributes significantly to enhance fair learning, providing support to researchers to direct their future works to the existing gaps, fostering a roadmap toward mitigating FL bias.

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